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High-Throughput Screening for the Development of a Monoclonal Antibody Affinity Precipitation Step Using ELP-Z Stimuli Responsive Biopolymers

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ABSTRACT: This study provides a detailed investigation into the performance of a stimuli responsive ELP-Z based process for monoclonal antibody (mAb) affinity precipitation. A multidimensional high-throughput screening (HTS) protocol was developed and employed to investigate the effects of a variety of operating conditions on mAb yield and aggregation during the process. Precipitation efficiency of ELP-Z in the absence of mAb was first determined as a function of temperature and sodium sulfate concentration and conditions producing high yields were identified. HTS was then employed to determine appropriate conditions for the initial capture and co-precipitation of mAbs at high yields using ELP-Z. mAb elution from ELP-Z was then examined using HTS and the mAb yields and aggregate content of the overall process were determined. It was observed that mAb aggregation was sensitive primarily to the elution conditions and that this behavior was antibody specific and a strong function of operating temperature and elution pH. Importantly, for both mAbs examined in this study, the results indicated that room temperature operation and appropriate elution pH could be readily employed to produce both high mAb yields and low aggregate content using this approach. This study demonstrates the ability of ELP-Z based affinity precipitation for mAb purification and shows that HTS can be successfully employed to rapidly develop a robust and high yield process.

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Introduction

Protein A affinity chromatography is widely employed for industrial antibody purification (Fassina, 2001; Hober et al., 2007; Huse et al., 2002). Recent years have seen a dramatic increase in mAb titers (5–10 g/L) (Huang et al., 2010) which has put significant pressure on the scale up and economics of the primary recovery step in downstream processing of mAbs. Further, volumetric throughput limitations have resulted in long processing times for Protein A column chromatographic operations. Accordingly, there has been significant interest in the development of non-chromatographic mAb recovery processes (Low et al., 2007; Thömmes and Etzel, 2007). One approach that has been explored is mAb precipitation using salts (Venkiteshwaran et al., 2008), polyelectrolytes (McDonald et al., 2009), and polyethylene glycols (Knevelman et al., 2010). While these approaches have been successful for recovery of the product, the purification factors achieved with these precipitants have not achieved the levels of Protein A. An alternative approach, affinity precipitation, combines the selectivity of affinity chromatography along with the scalability and cost benefits of precipitation (Hilbrig and Freitag, 2003).

Elastin-like polypeptides (ELPs) are stimuli responsive smart biopolymers that reversibly aggregate above a critical temperature called the inverse transition temperature (T_i) (Urry, 1997). ELPs are comprised of elastin-based pentapeptide repeating motifs (VPGXG), where X is the guest amino acid residue. The phase transition property of ELPs has been extensively studied as a function of ELP chain length and guest amino residue (Meyer and Chilkoti, 2004). In addition, the ELP phase transition is also affected by solution conditions such as pH, salt type, and concentration (Cho et al., 2008). These properties of ELPs have garnered interest in its potential for downstream bioprocessing applications (Banki et al., 2005; Kim et al., 2005; Meyer and Chilkoti, 1999; Stiborova et al., 2003; Sun et al., 2005).

Chilkoti and co-worker have demonstrated the applicability of ELP to purify target proteins fused directly to ELP

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(Meyer and Chilkoti, 1999). In that approach, the target protein is proteolytically cleaved from the ELP after the first precipitation step which is then followed by a second precipitation step to remove the cleaved ELPs. In an effort to eliminate the proteolytic cleavage step, Banki et al. (2005) have demonstrated the purification of proteins by fusing them to ELPs via a self-splicing intein domain. While the fusion approach is quite interesting, this approach often results in relatively low titers for the target protein, making it problematic for antibody purification in the current embodiment. Kim et al. (2005) have employed ELP-Protein A fusions as an affinity precipitating agent for mAbs. In this process, the mAb is first selectively captured from solution by the ELP-Protein A fusion, followed by precipitation of the complex. The precipitant is subsequently re-solubilized, the complex is dissociated and the ELP-Protein A fusion is precipitated leaving the mAb in the solution.

Methods development for affinity precipitation in an industrial setting can be challenging. In particular, for ELP based affinity precipitation, a variety of conditions such as temperature, salt type and concentration, pH, and other additives can affect the precipitation processes (Cho et al., 2008; MacKay et al., 2010; Meyer and Chilkoti, 2004) as well as the binding and elution of the antibody. In addition, the impact of these conditions on product quality and purity must also be determined. The development of a robust ELP based affinity precipitation process clearly requires a detailed process characterization.

In the present study, a high-throughput screening (HTS) strategy is developed and employed to fully characterize the performance of the ELP-Z antibody affinity precipitation process shown in Figure 1. The ELP-Z fusion employed in this work contains a single antibody binding Z domain along with an ELP consisting of a VPGVG pentapeptide repeat sequence with 78 repeats (Kim et al., 2005; Madan et al., 2013). HTS is carried out to examine the following: precipitation efficiency of the ELP-Z alone, initial capture and co-precipitation of the

mAb by ELP-Z, and mAb elution from the ELP-Z-mAb complex followed by a final ELP-Z precipitation step. This analysis includes an examination of the effect of a range of operating conditions on mAb yield and aggregation (a critical quality attribute) during each step of the process. The results of this study demonstrate how HTS can be successfully employed to rapidly develop an ELP-Z based affinity precipitation process for mAb purification.

Materials and Methods

Materials

A Jupiter 5 μm C4 300A column ($4.6 \times 50 \text{ mm}^2$) was purchased from Phenomenex (Torrance, CA). A TSKgelG3000SWxl SEC column ($7 \times 300 \text{ mm}^2$) with accompanying guard column was donated by Tosoh (Tokyo, Japan). V-bottomed 96 well assay plates were purchased from Costar (Corning, Corning, NY). 96 well plate sealing mats were purchased from Matrix (Thermo Fisher Scientific, Pittsburgh, PA). *Escherichia coli* strain BL21(DE3) cells containing the ELP-Z (ELP with valine as the guest amino acid residue and 78 pentapeptide repeats) plasmid were constructed as described (Kim et al., 2005). mAb A and mAb B were supplied by Bristol-Myers Squibb (BMS). Bacto tryptone and yeast extract were purchased from BD (Franklin lakes, NJ). Glycerol, ampicillin, 3-(*N*-morpholino)propanesulfonic acid (MOPS), sodium hydroxide and potassium chloride were purchased from Thermo Fisher Scientific. Sodium chloride, potassium phosphate (mono and dibasic), protease inhibitor cocktail, sodium sulfate, sodium citrate dihydrate, citric acid, sodium phosphate (dibasic), acetonitrile (ACN), trifluoroacetic acid (TFA), and arginine hydrochloride were purchased from Sigma-Aldrich (St. Louis, MO).

ELP-Z Expression and Purification

Escherichia coli strain BL21(DE3) containing the ELP-Z plasmid was grown in 3 mL Luria Broth (LB) with 100 $\mu\text{g}/\text{mL}$ ampicillin at 37°C and 250 rpm for 16 h. The grown cultures were then sub cultured into 250 mL Terrific Broth (100 $\mu\text{g}/\text{mL}$ ampicillin) and grown for 48 h at 37°C and 250 rpm. The final OD₆₀₀ was measured and the cells were harvested by centrifugation at 5,000g and 4°C for 15 min. The cell pellets were re-suspended (to OD₆₀₀ = 40) in phosphate-buffered saline (PBS) containing 1 $\times$ protease inhibitor cocktail (PBS; 137 mM NaCl, 2.7 mM KCl, 10 mM Na₂HPO₄, 2 mM KH₂PO₄, pH 7.4). The cells were disrupted by sonication (5 s pulse ON and 5 s pulse OFF for 10 min) and the cell debris was removed by centrifugation for 15 min and 4°C at 15,000g. The inverse phase transition property was then employed for the purification of the ELP-Z fusion. As mentioned elsewhere (Kim et al., 2005), NaCl was added to a final concentration of 1 M and the samples were heated to 37°C for 15 min followed by centrifugation at 5,000g and 37°C for 30 min. The pellets containing the ELP-Z fusion were dissolved in PBS at 4°C and centrifuged at 5,000g and

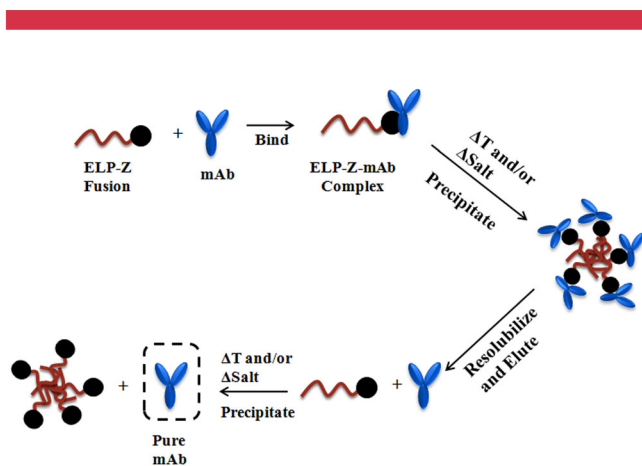


Figure 1. ELP-Z based mAb affinity precipitation strategy.

4°C for 30 min to remove any insoluble precipitates. Inverse transition cycling was repeated once more and the pellets containing ELP fusions were finally dissolved in ice-cold PBS. The purity of the ELP-Z was determined by SDS-PAGE and reversed phase liquid chromatography (RPLC) (as described below).

High-Throughput Screening for mAb Affinity Precipitation

The ELP-Z precipitation efficiency, mAb binding and precipitation, and the mAb elution experiments were all performed using 96 well v-bottom plates. For the ELP-Z precipitation efficiency experiments, in each well 100 µL of ELP-Z in PBS at 2.5 mg/mL was mixed with 25 µL of sodium sulfate in PBS at five times the desired final concentration (all the conditions were examined in triplicates). The 96 well plates were covered using sealing mats to minimize losses due to evaporation and were incubated in a water bath at the desired precipitation temperature (22, 30, and 37°C) for 15 min. The plates were then centrifuged at 3,000g for 45 min at the appropriate operating temperatures. After centrifugation, the supernatants were immediately collected using a multichannel pipettor. One hundred twenty microliters of the supernatants were collected carefully to minimize any solids removal. The supernatants were analyzed using RPLC to determine the amount of ELP-Z precipitated. In addition, the ELP-Z precipitates were also re-solubilized in 120 µL of PBS and analyzed using RPLC to calculate the overall mass balance.

For the mAb binding and precipitation studies, 50 µL of the mAb (in PBS) at 5 mg/mL was mixed with 50 µL of ELP-Z at different concentrations (in PBS) to achieve 4:1, 3:1, 2:1, and 1:1 ELP-Z:mAb molar ratios in the wells. The solutions were equilibrated for 10 min to allow ELP-Z–mAb binding. To initiate the precipitation of the ELP-Z–mAb complex, 25 µL of sodium sulfate (in PBS) at five times the desired final concentration was added to each well. The same procedure as described above was then carried out for plate incubation, centrifugation, and recovery of the supernatants. The precipitates were re-solubilized in PBS and also analyzed along with the supernatants for mAb and ELP-Z recoveries using RPLC.

For the mAb elution screens, the precipitates obtained from the mAb binding and precipitation step were re-solubilized in 100 µL of various elution buffers (50 mM citrate buffers between pH 3.6 and 5). The solutions were allowed to equilibrate at room temperature for 10 min. After elution was complete, the ELP-Z precipitation was initiated by adding 25 µL of sodium sulfate solutions (in the appropriate elution buffers) at five times the desired final concentrations to each well. The solutions were incubated at the desired operating temperatures in a water bath for 15 min followed by centrifugation at 3,000g for 45 min. The supernatants after centrifugation were collected using a multi channel pipettor and analyzed using size exclusion chromatography (SEC) for final mAb yields and aggregate content.

RPLC Analytical Technique

RPLC was used to analyze the mixtures (supernatants and re-solubilized precipitates) from the high throughput put experiments, to quantify the ELP-Z and mAb recoveries. RPLC was carried out using a C4 column (4.6 × 50 mm²) with an A buffer of deionized water with 0.1% TFA (v/v) and a B buffer of 90% ACN, 10% deionized water, and 0.1% TFA (all v/v). The column was first pre-equilibrated with 57% A and 43% B which caused the mAbs (mAb A and mAb B) to come out in the column flow through (monitored at 280 nm). This was followed by a step change to 100% B for two column volumes which resulted in the ELP-Z elution (monitored at 214 nm). The flow rate was 1 mL/min and 20 µL sample injections were made. Total analysis time was 7 min for this step gradient analysis.

SEC Analytical Technique

Size exclusion chromatography (SEC) was employed to analyze the final supernatants after the mAb elution screens to determine the mAb yields and aggregate content. A carrier buffer of 200 mM arginine with 20 mM MOPS, pH 6.5 was used at 1 mL/min flow rate. Forty microliters of sample injections were made and the column effluents were analyzed by UV/VIS at 280 nm. The samples were kept at 4°C in a temperature controlled auto injector (Waters 717plus Autosampler) to minimize aggregation of the product before analysis.

Results and Discussion

The affinity precipitation process is affected by the operating conditions required for ELP-Z precipitation, mAb-Z domain binding, and elution of the ELP-Z–mAb complex. The conditions needed for ELP-Z phase transition can also have an effect on the ELP-Z–mAb binding affinity. Importantly, the operating conditions employed at each step of the process can also impact the final product quality. In order to establish the effects of these various operating conditions on the overall process, the HTS approach shown in Figure 2 has been employed.

As can be seen in the figure, there are two major components of the process, (1) ELP-Z–mAb binding followed by precipitation of the complex and (2) resolubilization, elution of the ELP-Z–mAb complex, and precipitation of the ELP-Z fusion. HTS was employed for each step in the process. For a given ELP-Z fusion, precipitation was first studied under different temperature and solution conditions to establish initial operating ranges for precipitation of the ELP-Z fusion alone. This knowledge was then employed in concert with the established conditions for mAb–Protein A binding to screen a range of conditions for mAb recovery during this first binding/precipitation part of the process (note: the precipitation of the ELP-Z–mAb complex is not necessarily going to be the same as the precipitation of ELP-Z fusion alone). Once appropriate conditions were established, a HTS for the next phase of the process was initiated.

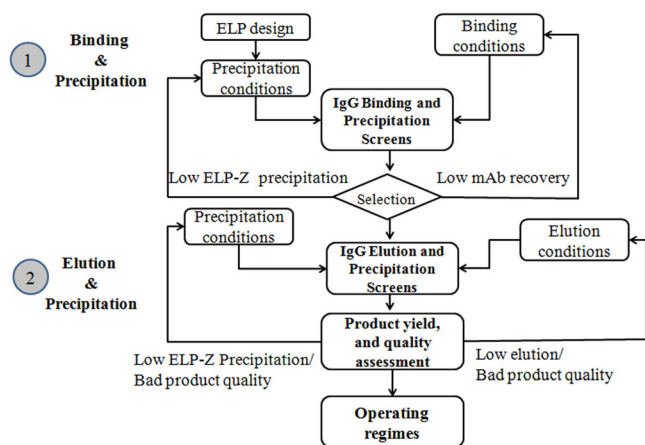


Figure 2. HTS approach for mAb affinity precipitation using ELP-Z smart biopolymer.

A variety of elution conditions were screened wherein the ELP-Z–mAb precipitates were initially resolubilized in the elution buffers. This was followed by the precipitation of the unbound ELP-Z fusion by the addition of salt and/or a temperature change. These set of screens were evaluated for mAb yield, ELP-Z precipitation, and mAb product quality (aggregation). If the desired outcome was not achieved due to improper elution or aggregate content, a new set of elution and precipitation conditions were evaluated. This process was repeated until the desired final outcome was achieved. Each step of the process will now be described in detail.

ELP-Z Precipitation Efficiency

Before carrying out the multidimensional HT screen, it was first important to examine the precipitation performance of the ELP-Z biopolymer under different temperatures and solution conditions. Sodium sulfate was used as the salt for the precipitation studies since it is a strong kosmotrope and has been shown previously to significantly affect the ELP phase transition at low concentrations (Cho et al., 2008). The results for ELP-Z precipitation at three different temperatures as a function of sodium sulfate concentrations are shown in Figure 3. As can be seen in the figure, at 22°C complete ELP-Z precipitation was achieved for 0.25 M and higher sodium sulfate concentrations. The precipitation efficiency reduced below 0.25 M sodium sulfate with no precipitation occurring at 0.125 M and lower concentrations. At 30°C, complete ELP-Z precipitation was achieved for 0.2 M and higher sodium sulfate concentrations. The precipitation efficiency gradually reduced between 0.2 and 0.1 M sodium sulfate followed by a steep decrease from 82% to 8% between 0.1 and 0.05 M sodium sulfate. No ELP-Z was precipitated at 0 M sodium sulfate at this temperature. At 37°C, complete ELP-Z precipitation was achieved for 0.1 M and higher sodium sulfate concentrations, whereas, the precipitation efficiency

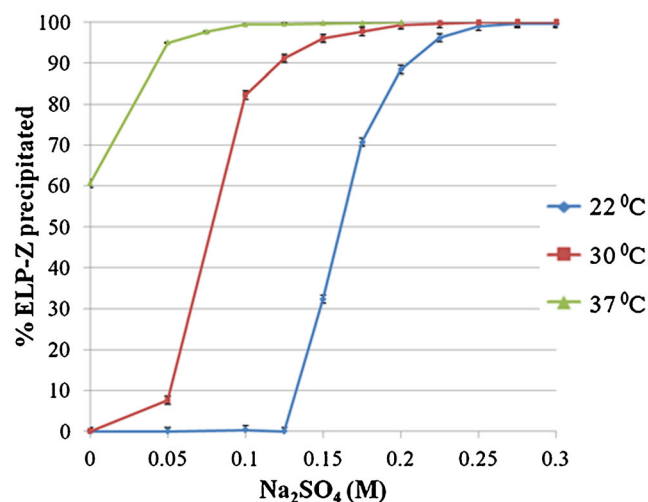


Figure 3. ELP-Z precipitation efficiency at different temperatures and sodium sulfate concentrations.

reduced below 0.1 M sodium sulfate to 60% at 0 M sodium sulfate.

These results for ELP-Z precipitation at a given temperature indicated that the precipitation efficiency reduced with decreasing sodium sulfate concentrations which was expected for this kosmotrope. The results for ELP-Z precipitation efficiency at different temperatures indicated that higher sodium sulfate concentrations were required to obtain complete ELP-Z precipitation at lower operating temperatures (0.1 M at 37°C, 0.2 M at 30°C, and 0.25 M at 22°C). This behavior was expected since as the salt concentration is reduced, the transition temperature of the ELPs should increase (Cho et al., 2008) which in turn should result in lower precipitation efficiency.

The different combinations of salt concentration and temperature for complete ELP-Z precipitation could also affect the binding and recovery of mAb by the ELP-Z construct. A subset of the conditions identified from the ELP-Z precipitation screen were then employed for determining mAb affinity precipitation performance of the ELP-Z construct, as described in the next section.

HTS for mAb Binding and Precipitation

As described in Figure 2, the first step of the multidimensional HTS screen was the evaluation of mAb binding and precipitation. A series of HTS experiments were performed to determine the mAb affinity precipitation performance of the ELP-Z construct. The binding and precipitation of the ELP-Z–mAb complexes were performed at three different temperatures (22, 30, and 37°C) using a range of sodium sulfate concentrations identified from the ELP-Z precipitation experiments described in Figure 3. The effect of using different ELP-Z:mAb molar ratios on the mAb precipitation

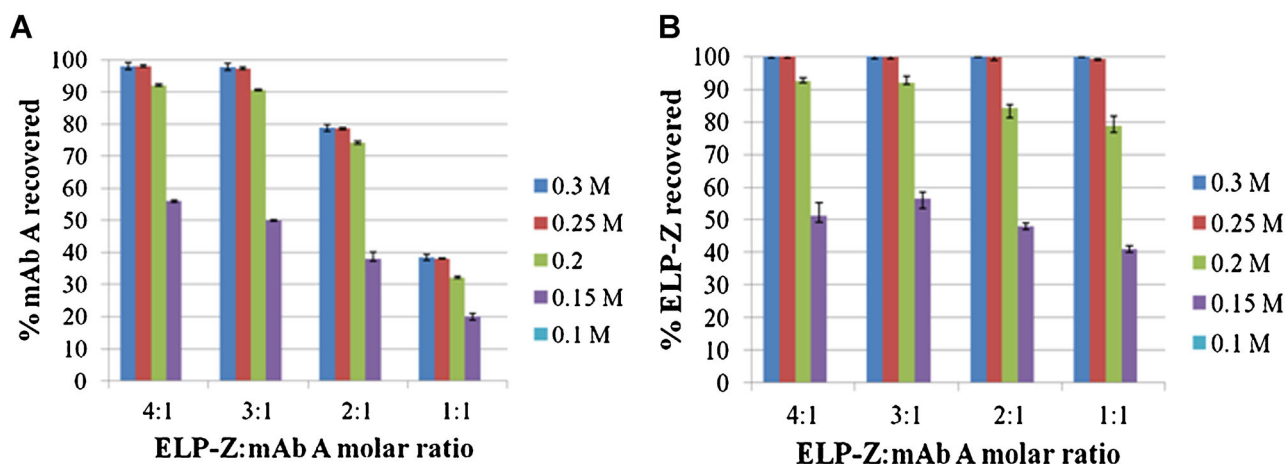


Figure 4. mAb A and ELP-Z recoveries after precipitation at 22°C, the legend indicates the sodium sulfate concentrations used for precipitation. **A:** mAb A recoveries after precipitation and **(B)** ELP-Z recoveries after precipitation.

was also determined. Figure 4A shows the results for mAb A affinity precipitation at 22°C. As can be seen in the figure, at the higher salt concentrations high mAb A recoveries (close to 98%) were obtained at 4:1 and 3:1 ELP-Z:mAb A molar ratios whereas only 79% and 37% mAb A was recovered at 2:1 and 1:1 ratios, respectively. At each ELP-Z:mAb A ratio, the mAb A recoveries were similar for 0.3 and 0.25 M sodium sulfate whereas the recoveries were reduced at the lower salt concentrations.

It was also of interest to examine the ELP-Z recoveries obtained during these precipitation experiments with the ELP-Z–mAb A complex (Fig. 4B). As can be seen from the figure, almost complete ELP-Z was recovered at all ratios for 0.3 and 0.25 M sodium sulfate concentrations. At 0.2 M sodium sulfate concentration, the ELP-Z recovery reduced to 91% for the 4:1 and 3:1 ratios and 84% and 79% for 2:1 and 1:1 ratios, respectively. The ELP-Z recoveries significantly reduced at 0.15 M sodium sulfate and no ELP-Z was recovered at 0.1 M sodium sulfate. These results indicate that the salt concentration affected the recovery of the ELP-Z in the presence of the mAb A at all ratios. Interestingly, the recovery of the ELP-Z was in general greater than the mAb A recovery under most conditions.

The results shown in Figure 4A indicate that ELP-Z:mAb A molar ratios of 3:1 or higher were required to obtain high levels of mAb A recovery during the precipitation at 22°C. The stoichiometry of binding of the Z domain to mAb should be on the order of 2:1 (Jendberg et al., 1995). There are two possible explanations for why a higher molar ratio is required for good mAb recovery. One possible explanation for this behavior could relate to the difference in molecular weights of the ELP-Z and mAb A. The molecular weight of mAb A (around 150 kDa) is much higher than that of ELP-Z (around 40 kDa). Since ELP-Z is the protein that self associates during precipitation, this difference in molecular weights may add

significant steric barriers to the ELP precipitation. Thus, higher quantities of ELP-Z may be required to facilitate the co-precipitation of the mAb after it is bound to the ELP-Z through the use of “helper” ELP-Z precipitation agents which do not participate in direct binding to the mAb.

Another explanation relates to the binding affinity of this ELP-Z to the mAb. The binding affinity of the Z domain to mAb A may be reduced due to fusion with the ELP. The ELP could sterically interfere with the Z domain binding to mAb A since it is a flexible polymer and the Z (58 amino acids) domain is much smaller as compared to the ELP (390 amino acids). Such a reduction in binding affinity on fusion to smart polymers has been previously reported for a Protein A—pNIPAM system (Chen and Huffman, 1990). If the binding affinity is not sufficiently strong, there may be some dissociation of the complex during the precipitation process, necessitating a higher molar ratio. In fact, this is supported by the data in Figure 4B which indicates that good recovery of the ELP-Z was achieved at the lower molar ratios whereas a significant reduction in mAb A recovery occurred. If all of the ELP-Z remained bound to the mAb A, the recoveries should have been comparable. This indicates that the ELP-Z was either not bound to the mAb A under solution conditions or that it dissociated during the precipitation process due to the lower binding affinity.

Results for mAb A affinity precipitation at 30°C are shown in Figure 5. Figure 5A shows the mAb A recovery as a function of ELP-Z:mAb A molar ratios and the precipitation salt concentrations. High levels of mAb A recoveries were obtained at 4:1 and 3:1 ELP-Z:mAb A molar ratios and the recoveries reduced to 76% and 38% at 2:1 and 1:1 molar ratios at the higher salt concentrations. For a given ELP-Z:mAb A molar ratio, the mAb A recoveries were similar for 0.25 and 0.2 M sodium sulfate concentrations with a gradual decrease in recovery for 0.15 and 0.1 M sodium sulfate

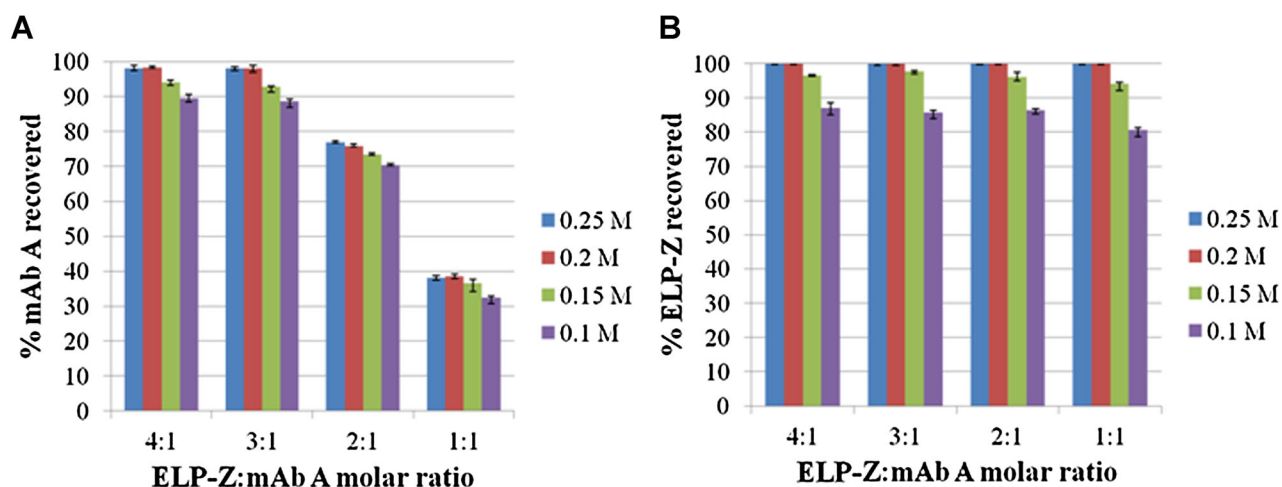


Figure 5. mAb A and ELP-Z recoveries after precipitation at 30°C, the legend indicates the sodium sulfate concentrations used for precipitation. **A:** mAb A recoveries after precipitation and **(B)** ELP-Z recoveries after precipitation.

concentrations. However, the salt dependence was much less pronounced at 30°C as compared the results for 22°C. The ELP-Z recoveries at 30°C are given in Figure 5B which shows that these recoveries were also much less salt dependent as compared to 22°C and that they were essentially molar ratio independent. Again, the recoveries for the mAb A were less than those obtained for the ELP-Z at the lower molar ratios.

Figure 6 shows the results for mAb A affinity precipitation by ELP-Z at 37°C. Again, high levels of mAb A recoveries (Fig. 6A) were obtained at 4:1 and 3:1 ELP-Z:mAb A molar

ratios, whereas lower recoveries were obtained at 2:1 and 1:1 ratios. mAb A recoveries were similar for 0.15 M and 0.1 M sodium sulfate concentrations with a gradual decrease for 0.05 M and 0 M sodium sulfate concentrations. The ELP-Z results at 37°C are given in Figure 6B which shows very similar results to the 30°C experiments.

Upon comparing the results shown in Figures 4–6, it can be concluded that at least 3:1 ELP-Z:mAb A molar ratios were required to obtain high levels of mAb A recovery in the first precipitation step of the process. Interestingly, at lower

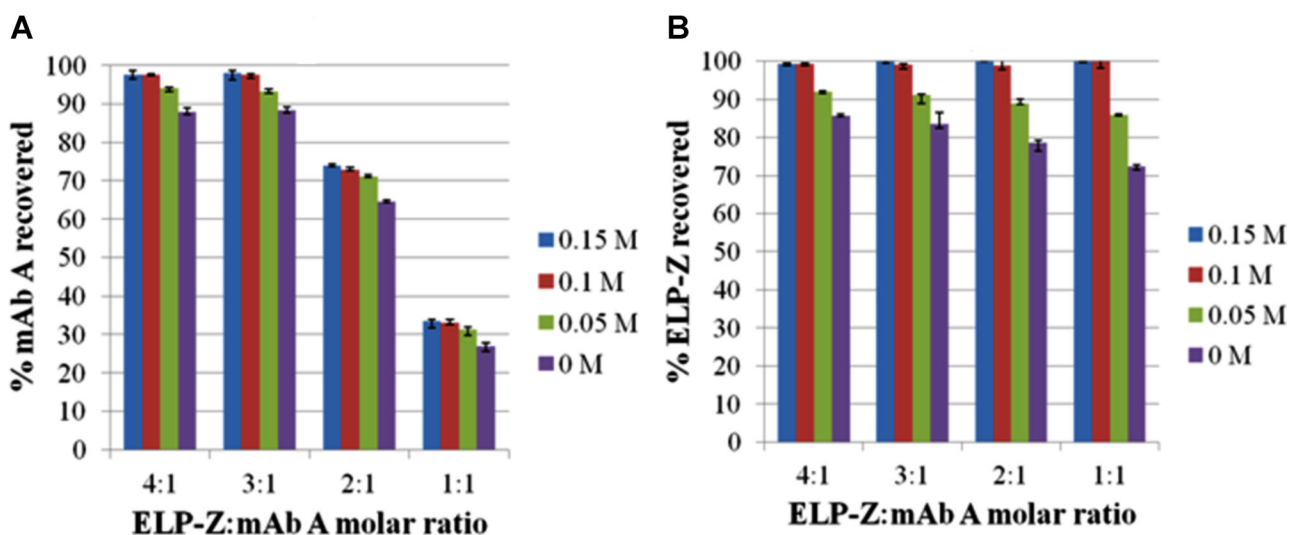


Figure 6. mAb A and ELP-Z recoveries after precipitation at 37°C, the legend indicates the sodium sulfate concentrations used for precipitation. **A:** mAb A recoveries after precipitation and **(B)** ELP-Z recoveries after precipitation.

ELP-Z:mAb A molar ratios complete ELP-Z recoveries were obtained despite lower mAb A recoveries, even at high salts. The salt concentrations required for high levels of precipitation reduced with increasing operating temperature, which was expected from the ELP-Z precipitation results. Additionally, at higher temperatures, the ELP-Z and the mAb A recoveries became less salt sensitive. Finally, for a given ELP-Z:mAb A molar ratio, the operating temperature by itself did not significantly affect the mAb A recovery, given that appropriate salt concentration was used for the precipitation.

Similar trends were obtained for the affinity precipitation of mAb B with the ELP-Z smart biopolymer where good recovery of mAb B was achieved at ELP-Z:mAb B molar ratios of 3:1 and above. The optimum operating conditions identified from this first precipitation step were then employed in the next stage of the HTS screen to investigate mAb elution from the ELP-Z–mAb complex.

HTS for mAb Elution: mAb Yields

In this stage of the process, a multidimensional HTS was employed to determine the elution of the mAbs from the ELP-Z–mAb complex. The mAbs were initially precipitated using the optimized conditions from the first step of the process described above. The ELP-Z–mAb precipitates were then re-solubilized in different elution buffers to examine the efficacy of the elution and the quality of the product. This was then followed by another round of ELP-Z precipitation. The binding of the Z domain to the antibodies is extremely sensitive to pH and low pH is commonly employed for mAb elution from Protein A based resins (Ghose et al., 2005). Accordingly, a number of low pH conditions were employed for screening the mAb elution. Fortunately, since the ELP domain employed for precipitation has no ionizable groups, its inverse transition properties are relatively insensitive to operating pH thus enabling precipitation conditions identified from the first precipitation step (temperature and salt concentration) to be directly employed for the second precipitation step. The mass balances were carried out for these experiments and >98% mass balance was generally achieved.

Figure 7 shows the results for mAb A yields after elution as a function of the elution pH, operating temperatures, and precipitation salt concentrations. It is important to note that this data on mAb yields represent the overall process yields that include the first precipitation step, resolubilization, elution, and the second precipitation step. However, since the first precipitation step of the process provided almost complete mAb recovery, the results shown in Figure 7 primarily reflect the mAb A yields after the final elution/precipitation step. Figure 7A shows the mAb A yields at 37°C. As can be seen in the figure, approximately 90% mAb A yield was achieved up to pH 4.2 at all the salt concentrations tested. The yields reduced to close to 80% at pH 4.5 and below 30% at pH 5 for all the salt concentrations tested (note: pH 3.6 was not used for elution since the mAb A produced significant turbidity at this pH at 37°C). Figure 7B shows the mAb A

yields at 30°C. As can be seen in the figure, close to 90% yields were obtained up to pH 4.2 for all the precipitation salt concentrations used. Again, there was a decrease in the mAb A yields at pH 4.5 and pH 5 (note: at this temperature, pH 3.6 results were also included since the antibody did not produce measurable turbidity). The results at 22°C were similar (Fig. 7C) with high yields obtained up to pH 4.2 followed by a decrease in the yields at pH 4.5 and 5 at all the precipitation salt concentrations examined.

HTS was also employed to examine the yield and aggregation occurring during mAb B elution from the ELP-Z fusion. Figure 8A shows the mAb B yields at 37°C. As can be seen in the figure, high mAb B yields (90–95%) were achieved up to pH 4.2. A small decrease in yield to 85% occurred at pH 4.5 and a sharp decrease in yield occurred at pH 5 (30%). Results at 30 and 22°C showed a similar trend, with high mAb B yields achieved up to pH 4.2 followed by a decline in yields for pH 4.5 and 5. The precipitation salt concentrations used in these screens were seen to have a negligible effect on the elution yields for the ranges employed. The yield results with mAb B elution were similar to the results obtained with mAb A (Fig. 7), although slightly higher mAb B yields were achieved.

The results shown in Figures 7 and 8 suggest that in general, high mAb yields (approximately 90%) were obtained at these three operating temperatures up to pH 4.2 followed by a slight decrease between pH 4.2 and 4.5, followed by a marked decrease between pH 4.5 and 5. At a given temperature, the precipitation salt concentration had a negligible impact on mAb yields. Finally, at a particular pH, operating temperature had minor effects on the mAb yields, again since this was dominated by the elution from the ELP-Z–mAb complex.

The high mAb yields at pH 4.2 are interesting since Z domain based affinity chromatographic systems typically require elution pH between 3.8 and 4 for high yields (Ghose et al., 2005). These resins (MabSelect SuRe, GE Healthcare, Uppsala, Sweden) contain a tetrameric engineered Z domain as the ligand that can exhibit higher binding affinity to the mAbs (Jendeborg et al., 1995) and thus require lower pH for elution. The ELP-Z fusion on the other hand contains only a single Z domain for mAb binding, which may be responsible for the milder pH elution conditions.

HTS for mAb Elution: Product Quality

Since aggregation is a critical quality attribute for mAbs, mAb A aggregation after elution was examined from the HTS experiments using SEC. Figure 9 shows the results for mAb A aggregation after elution. As can be seen in the figure, at 37°C (Fig. 9A), high levels of mAb A aggregation (40–50%) occurred at all salt concentrations at pH 3.8. This was followed by a sharp decrease in aggregation at pH 4 (10–15%). At pH 4.2 the extent of aggregation was approximately 2% which was equal to the native value (note: native value corresponds to the aggregates present in the mAb stock solution used for the experiment). At pH 4.5 and pH 5 the

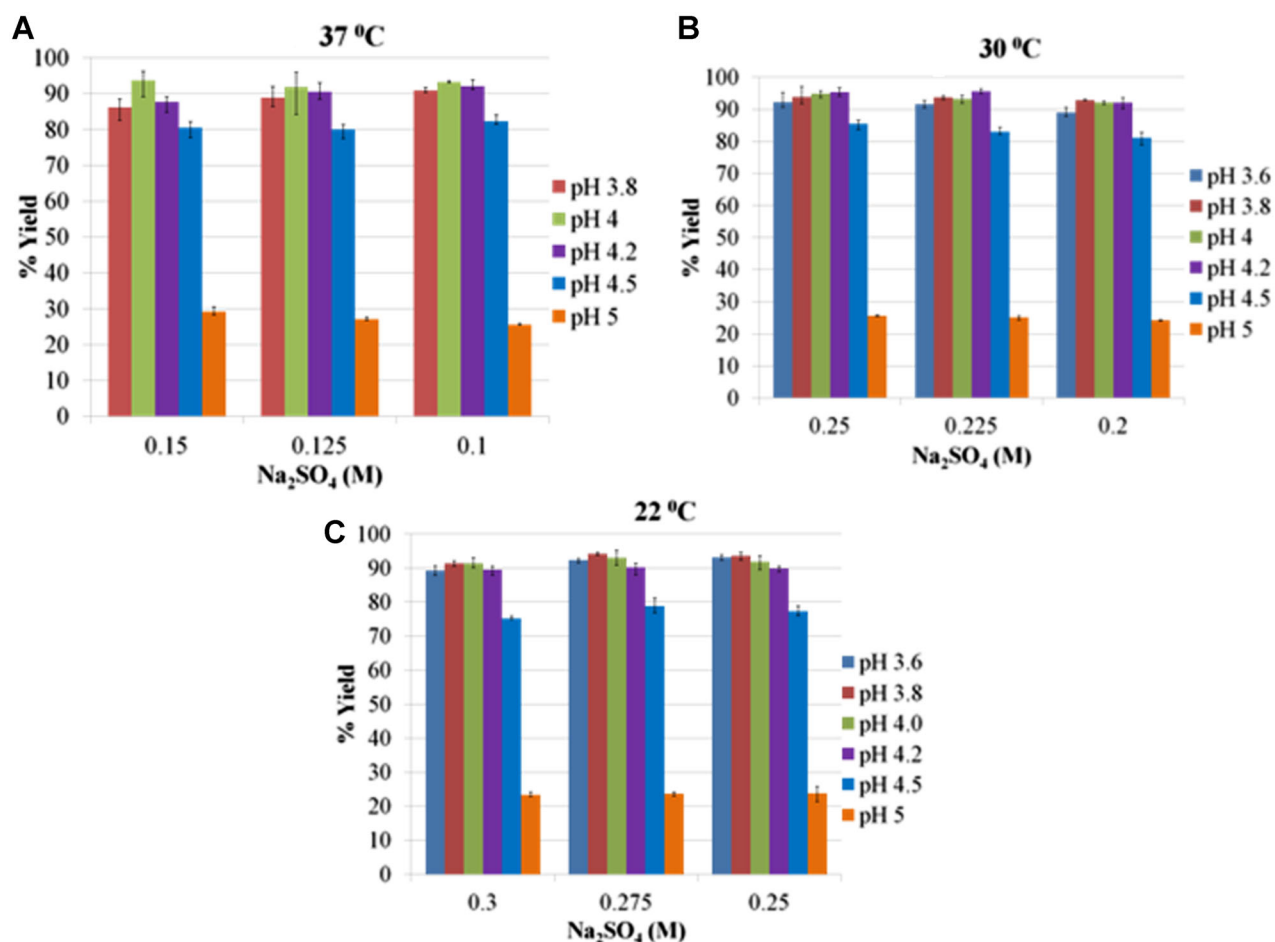


Figure 7. mAb A yields after elution from the ELP-Z and the second round of precipitation. **A:** mAb A yields at 37°C precipitation temperature. **B:** mAb A yields at 30°C precipitation temperature. **C:** mAb A yields at 22°C precipitation temperature.

extent of aggregation was below the native value. For all of these experiments, mAb A aggregation was seen to slightly reduce with decreasing salt concentration. These results suggest that the mAb aggregation was extremely pH sensitive during this process.

At 30°C (Fig. 9B), the extent of aggregation was lower than that obtained at 37°C. Under these conditions, pH 3.6 resulted in an aggregate content of 20–25% and pH 3.8 resulted in approximately 7% aggregates. At pH 4 and 4.2, the extent of aggregation was close to the native value whereas at pH 4.5 and 5 values lower than the native value were observed. Again, there was a slight increase in the aggregate content with increasing salt concentrations.

Finally, at 22°C (Fig. 9C), the extent of mAb A aggregation was significantly reduced as compared to the 30 and 37°C experiments. In fact, the aggregate content was below 6% for all the conditions examined. At elution pH of 3.8, 4, and 4.2 the mAb aggregation was close to the native value and at pH 4.5 and 5 it was below the native level as was observed for 30°C. For the 22°C experiments, the precipitation salt concentration had a negligible effect on mAb aggregation.

The results shown in Figure 9 suggest that mAb aggregation during the elution/precipitation process was a strong function of both the operating temperature and the elution pH. At a given temperature the extent of aggregation decreased with increasing elution pH and at a given pH, mAb aggregation decreased with decreasing temperature. In all of these product quality evaluations, the precipitation salt concentrations had a minimal impact on mAb A aggregation for the ranges employed. These results in comparison with the results from Figure 7 suggest that although the performance of this process at different temperatures was similar with respect to the mAb A yields, mAb A aggregate content after elution was significantly affected by the operating conditions. At 37°C, only pH 4.2 provided good mAb A yields with low aggregate content while at 30°C both pH 4 and 4.2 provided good product yields and low aggregation. Importantly, when the experiments were carried out at 22°C, elution pH between 3.8 and 4.2 all provided high yields and low concentrations of aggregates.

(Hari et al., 2010) have demonstrated that elevated sodium chloride concentrations along with low pH can induce mAb

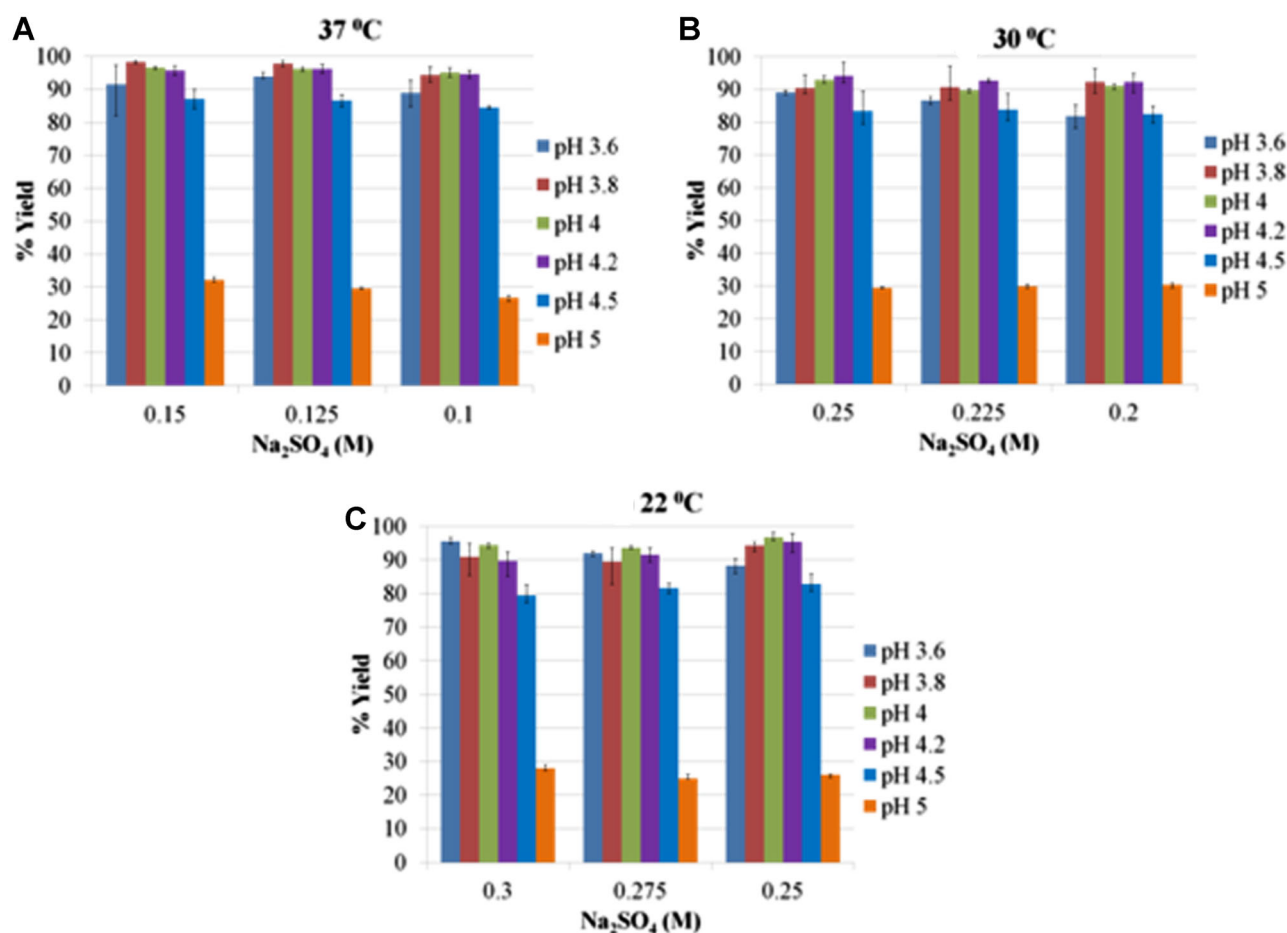


Figure 8. mAb B yields after elution from the ELP-Z and the second round of precipitation. **A:** mAb B yields at 37°C precipitation temperature. **B:** mAb B yields at 30°C precipitation temperature. **C:** mAb B yields at 22°C precipitation temperature.

aggregation. In order to investigate if the solution conditions employed during the elution/precipitation step were responsible for aggregation in the absence of ELP-Z and precipitation, a control experiment was carried out. mAb A was incubated at 37°C at pH 3.8 and 4.0 at sodium sulfate concentrations of 0 and 0.15 M. As can be seen in Figure 10, sodium sulfate concentration and pH had a dramatic effect on mAb A aggregation. In the presence of 0.15 M sodium sulfate, the extent of aggregation at both pH 3.8 and 4.0 were essentially the same as those observed during the actual affinity precipitation process (Fig. 9A). Similar results were obtained at other conditions as well. These results suggest that it was the solution phase conditions that were in fact responsible for the aggregation observed during the elution/precipitation step of the process. Importantly, precipitation did not appear to be a contributing factor to the mAb aggregation, but it was the salt and temperature conditions that were required to precipitate the ELP-Z fusion after elution that were the cause of the aggregation. The balance between mAb yield and low aggregate content will be explored in detail below.

mAb B aggregation after elution was also examined from the HTS experiments using SEC (Fig. 11). As can be seen in the figure, aggregation was significantly more pronounced for mAb B as compared to mAb A. At 37°C (Fig. 11A), very high levels of mAb B aggregation were observed at pH 3.6 and 3.8 (70–75%) as well as at pH 4 (50–60%). This was followed by a sharp decrease in aggregation at pH 4.2 (6–10%). At pH 4.5 and 5, mAb B aggregation was observed to be below the native value. At 30°C (Fig. 11B), mAb B aggregation was again quite high at pH 3.6 (47–50%) and pH 3.8 (27%), and reduced to close to the native value above pH 4. Finally, at 22°C, only the pH 3.6 condition resulted in elevated mAb B aggregate content (~19%), with the other pHs resulting in minimal aggregation. In all of these experiments, sodium sulfate concentration was not observed to have a measurable impact on mAb B aggregation for the ranges examined. pH sensitivity of the mAb B aggregation was observed to be to different at various temperatures. At 37°C, the aggregation was relatively insensitive to pH up to 4.0, after which there was a dramatic reduction in aggregate content going from pH 4 to 4.2. On the other hand, at 30°C, a continual reduction in

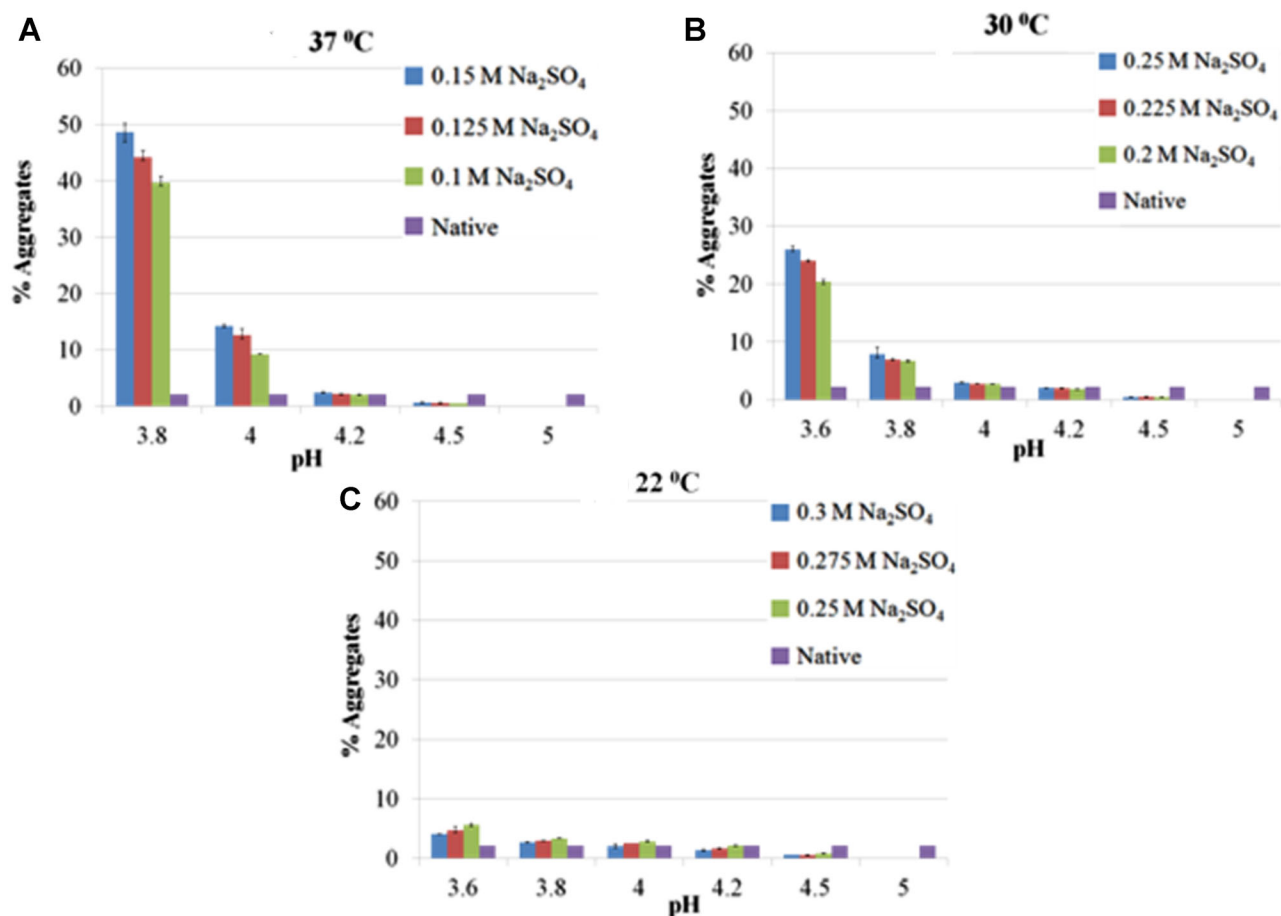


Figure 9. mAb A aggregation after elution from the ELP-Z. **A:** mAb A aggregation during precipitation at 37 °C. **B:** mAb A aggregation during precipitation at 30 °C and **(C)** mAb A aggregation during precipitation at 22 °C.

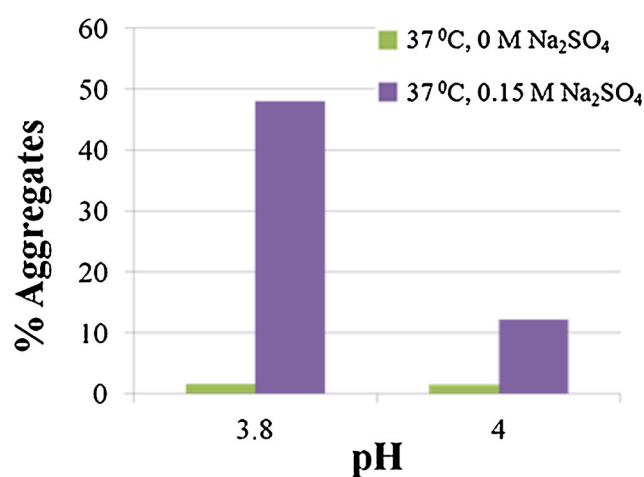


Figure 10. mAb A aggregation control experiment. The mAb A samples were incubated under these conditions for the same duration as during the precipitation experiments (total 60 min).

aggregate content was observed when going from pH 3.6 to 4.0. These results suggest that lower operating temperatures not only resulted in reduced mAb B aggregate content but also a potentially more robust process.

Clearly, these results indicate that the behavior of these two antibodies while similar with respect to yield, were quite different with respect to aggregation. These differences arise from the individual mAb's propensity to aggregate under the conditions employed. The differences in mAb aggregation behavior during the process can significantly affect the safe operating regimes for a robust process and demonstrate the need for HTS to identify these regimes.

In order to compare the elution performance of the two mAbs, Figure 12 shows a 2D heat map analysis for both mAb yields and aggregation as a function of the operating temperatures and elution pH (note: since the optimized salt concentration range used for the screens had minimal effect on the process performance, a single salt concentration was employed for this analysis). Figure 12A and C present the yield maps for mAb A and mAb B, respectively, and Figure 12B and D show the aggregation heat maps for these

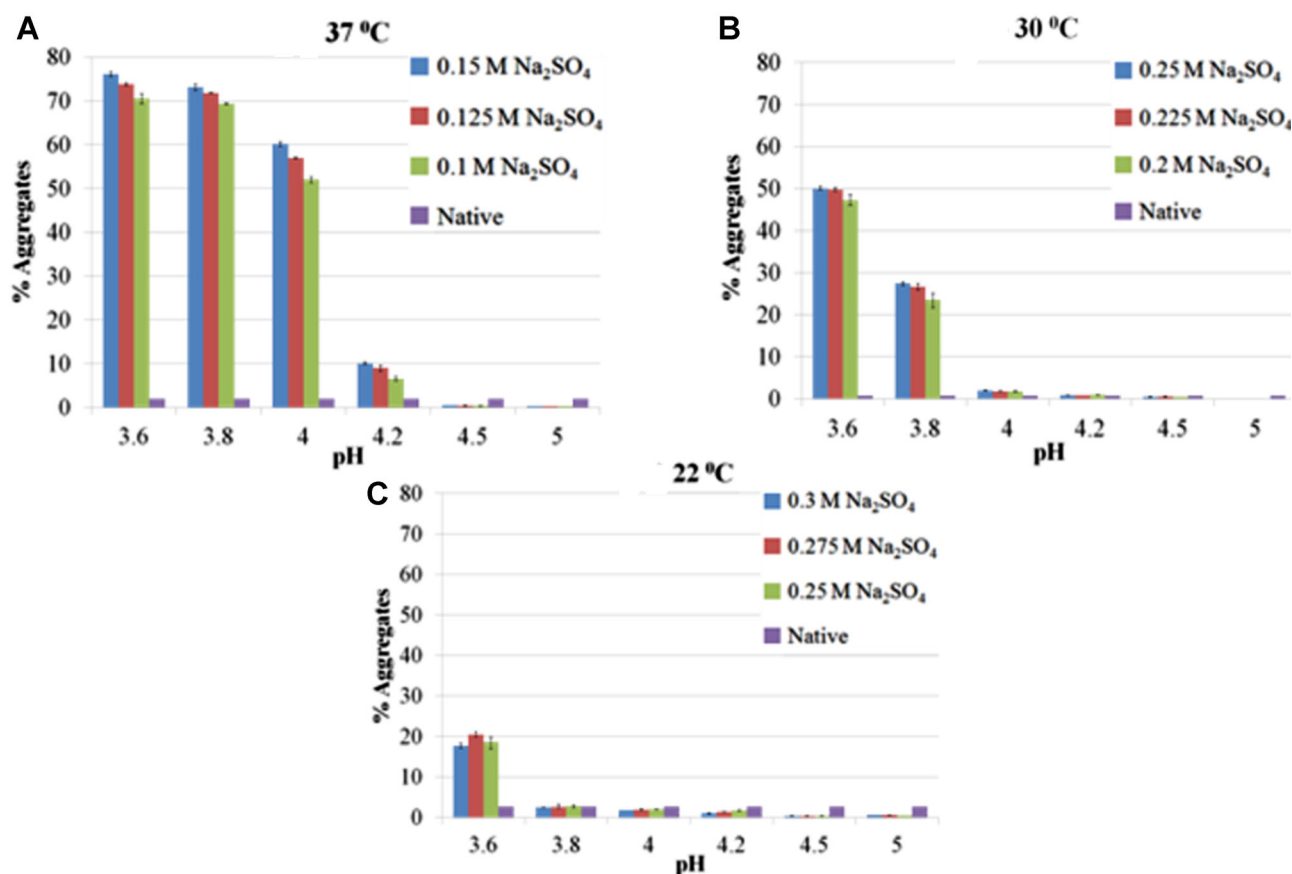


Figure 11. mAb B aggregation after elution from the ELP-Z. **A:** mAb B aggregation during precipitation at 37°C. **B:** mAb B aggregation during precipitation at 30°C and **(C)** mAb B aggregation during precipitation at 22°C.

antibodies. Upon comparing the final process yields for mAb A (Fig. 12A) and mAb B (Fig. 12C), it can be seen that high overall mAb yields were obtained for both antibodies at pH 4.2 and below (note: pH 3.6 results are not included for mAb A at 37°C since this antibody produced a turbid solution under this condition). Further, operating temperature had a minimal effect on the yields for both the antibodies. The acceptable operating conditions for each antibody with respect to product yields are indicated in Figure 12A and C using the dashed lines and arrows.

In contrast to the mAb yields, the aggregation results for mAb A (Fig. 12B) and mAb B (Fig. 12D) were quite different. Both the operating temperature and the elution pH had a significant effect on the aggregate contents. As described above, mAb B was more aggregation prone than mAb A under the conditions employed for the elution screens. The acceptable operating ranges for each antibody with respect to aggregate content are indicated in the figures using dashed curves and upward arrows.

The final acceptable operating regime for each antibody is then determined by the overlap of the conditions providing high yield and low aggregation. This is indicated in Figure 12B and D by the region encompassed by the both

the dotted line corresponding to the yield requirement and the dashed curve corresponding to the aggregate content. For mAb A, at higher temperatures (e.g. 37°C), the operating pH range is very narrow. As the temperature decreases, the operating pH range for mAb A becomes wider and at 22°C a broad pH operating range is established. For mAb B, it is not possible to operate at higher temperatures because the operating regimes for high yield and low aggregation do not overlap. In fact, it is only possible to operate this process at moderate (30°C) and low operating temperatures (22°C). While operation at 22°C resulted in the widest elution pH range for mAb B, it was still more constrained than for mAb A. These elution results suggest that the aggregation propensity of each individual mAb will play a critical role in the determination of suitable operating regimes for affinity precipitation of a given mAb.

Conclusions

A multidimensional HTS strategy was employed to investigate the performance of an ELP-Z smart biopolymer system for affinity precipitation of mAbs and to examine the effects of a variety of operating conditions on product yield and

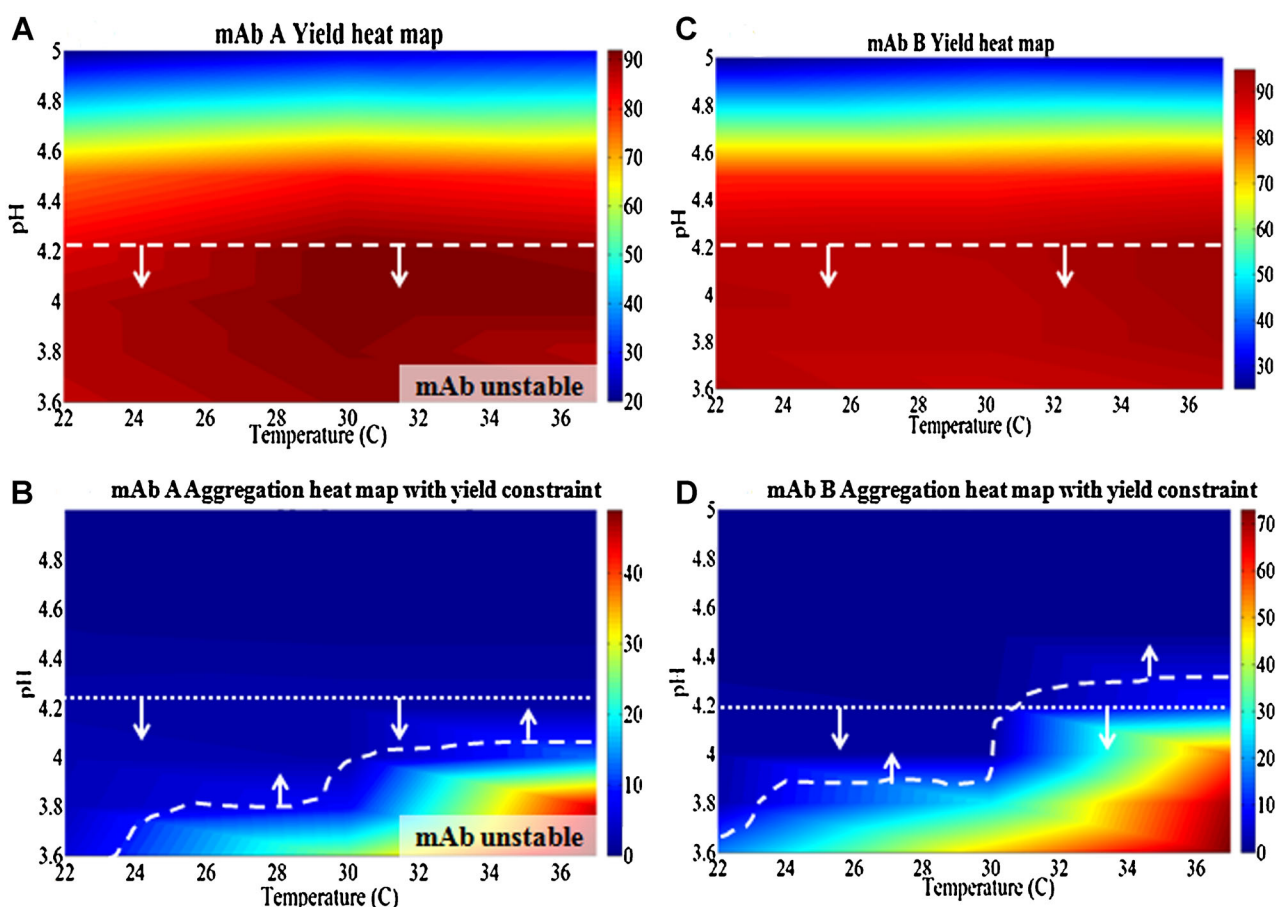


Figure 12. mAb A and mAb B final yield and aggregation 2D heat maps for the affinity precipitation process. **A:** mAb A yield heat map, **(B)** mAb A aggregation heat map, **(C)** mAb B yield heat map, and **(D)** mAb B aggregation heat map. The suitable operating conditions are shown in the figure using dotted lines and curves with the appropriate arrows. The overlap of high yield and low aggregation operating regimes are also shown in (B) and (D).

aggregation. For the first precipitation step, high mAb yields were achieved at ELP-Z:mAb molar ratios $\geq 3:1$. Lower salt concentrations were required to achieve high yields in this step at higher operating temperatures.

Antibody yields after elution in the second step of the process were primarily governed by the elution pH, where high yields were obtained up to pH 4.2 for both the mAbs examined. While high yields were obtained for a range of temperature and salt concentrations, mAb aggregation was strongly dependent on the operating temperature as well as the elution pH. The extent of aggregation and its sensitivity to the operating conditions was found to be quite different for the two mAbs examined, illustrating the importance of carrying out an appropriate screening protocol for each mAb product. Suitable operating regimes providing high mAb yields with low levels of aggregation were successfully identified for both mAbs using the HTS approach. In general, operation at room temperature (22°C) resulted in a wider operating elution pH range for both the mAbs.

This work provides a framework for developing ELP smart biopolymer based affinity precipitation processes for mAb

purification. The HTS approach employed here elucidates the effects of a multitude of operating conditions on the process performance and can be readily employed for a variety of mAbs. Further, while this work has focused on ELP based affinity precipitation for mAbs, the approach can potentially be employed for a range of bioproducts. Future work will focus on applying this technology to complex industrial mixtures and will examine process scale-up challenges and opportunities.

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